2024239

B.E. I YEAR EXAMINATION JANUARY 2024
1ETRC4- Basic Electronics
Information Technology (A &B) **M. Marks: 60**

Note: Attempt any two parts from each question.

Q1. a) How does the applied voltage across a PN junction affect the Fermi energy levels and the depletion region width? [6]

b) Compare and contrast the impacts of temperature on the carrier mobility in intrinsic and extrinsic semiconductors, explaining the underlying mechanisms. [6]

c) (i) Discuss the components required to obtain regulated DC power supply. [3]

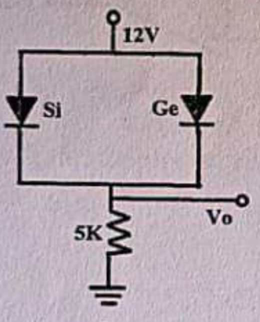
(ii) Calculate the reverse saturation current of a diode if the current at 0.2V forward bias is 0.1mA at a temperature of 25°C and the ideality factor is 1.5. [3]

Q2. a) Find RMS value, Average value, Ripple factor, Efficiency and PIV of half wave rectifier. Also, compare the performance of half wave rectifier with and without filter. [6]

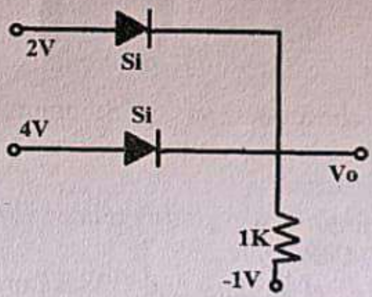
b) Draw and explain VI characteristic of tunnel diode using fermi energy level diagram. [6]

c) (i) Draw the ideal characteristics of diode and explain how it differs from that of a practical diode. [2]

(ii) From the circuit diagram given in figure (i) and (ii) find Vo. [4]



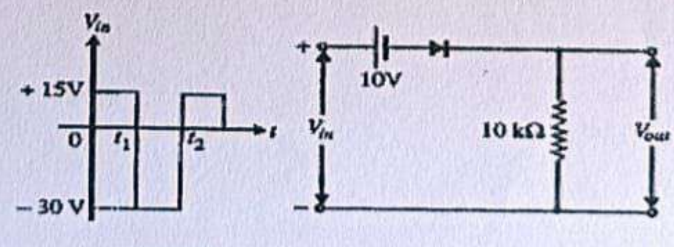
(i)



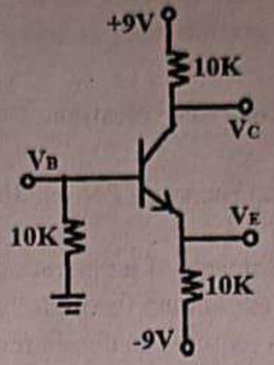
(ii)

Q3. a) Explain the working principal of NPN transistor. Also drive relations between α, β and γ (where α, β and γ are current gain in different transistor configurations). [6]

b) For the circuit and input waveform given below, (i) Draw output waveform (ii) Find the peak value and expression for the output. [6]



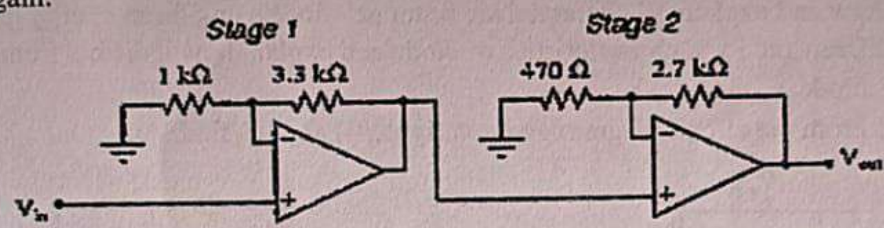
c) For the circuit shown below, measurement indicates the value V_B= -1.5 V, Assuming that V_{BE} =0.7 V, Calculate V_E, β and V_C. [6]



Q4. a) Explain the Ideal characteristics of Op-Amp (Operational Amplifier). [6]

b) What do you mean by differential amplifier and Push-Pull Amplifier? Explain mechanism with circuit diagram. [6]

c) Calculate the voltage gain for each stage of this amplifier circuit then also calculate the overall voltage gain. [6]



Q5. a) Write short notes on: (i) Summer Amplifier (ii) Difference Amplifier [6]

b) Describe logarithmic Amplifier and Anti-logarithmic Amplifier with circuit diagram and output expressions. [6]

c) Explain following terms related to Op-Amp: [6]

(i) CMRR (ii) Slew rate

(iii) Offset Voltage and Offset Current (iv) Level Shifter